ES32079 Financial Derivatives
Problem Set 2

Question 1

Differentiate between forward, futures, and options contracts in terms of:

- Standardization
- Settlement
- Credit risk
- Typical market participants

Question 2

A manufacturing company uses large quantities of copper in its production process and is concerned about rising copper prices over the next six months.

- What derivative instrument could the company use to hedge this price risk?
- Would it take a long or short position?
- Identify one potential risk that remains even after hedging.

Question 3

A non-dividend-paying stock is currently priced at \$50. The risk-free rate is 6% per annum, with continuous compounding.

- Calculate the 6-month forward price for the stock.
- If the time to maturity were instead 1 year, what would the forward price be?

Question 4

Explain briefly why the forward price may differ from the expected future spot price. Which forces ensure that the forward price is not arbitrageable?

Question 5

The current futures price for a Treasury bond deliverable in 3 months is 101.50. The bond's coupon is 6% per annum, with semiannual payments, and the next coupon is due in 4 months. The risk-free rate is 5% (continuous compounding).

- Explain qualitatively what factors might cause the current futures price (101.50) to be higher or lower than the current spot price of the bond.
- How might the timing of the next coupon payment affect the relationship between the spot price and the futures price?
- If the bond were zero-coupon, would you expect the futures price to be higher or lower than the spot price? Briefly explain.

Question 6

Define the following terms clearly:

- Spot rate
- Forward rate
- Par yield
- Zero-coupon rate

Briefly describe how these rates differ from one another in interpretation or use.

Question 7

You observe the following Treasury bond data:

Maturity (years)	Coupon Rate (annual)	Price per 100 par
0.5	0%	97.00
1.0	6%	101.50
1.5	8%	104.00

Assume annual coupon payments.

- Compute the 0.5-year zero rate.
- Using that, estimate the 1-year zero rate.
- Briefly explain what the next step would be if we wanted to extend the zero curve to longer maturities.